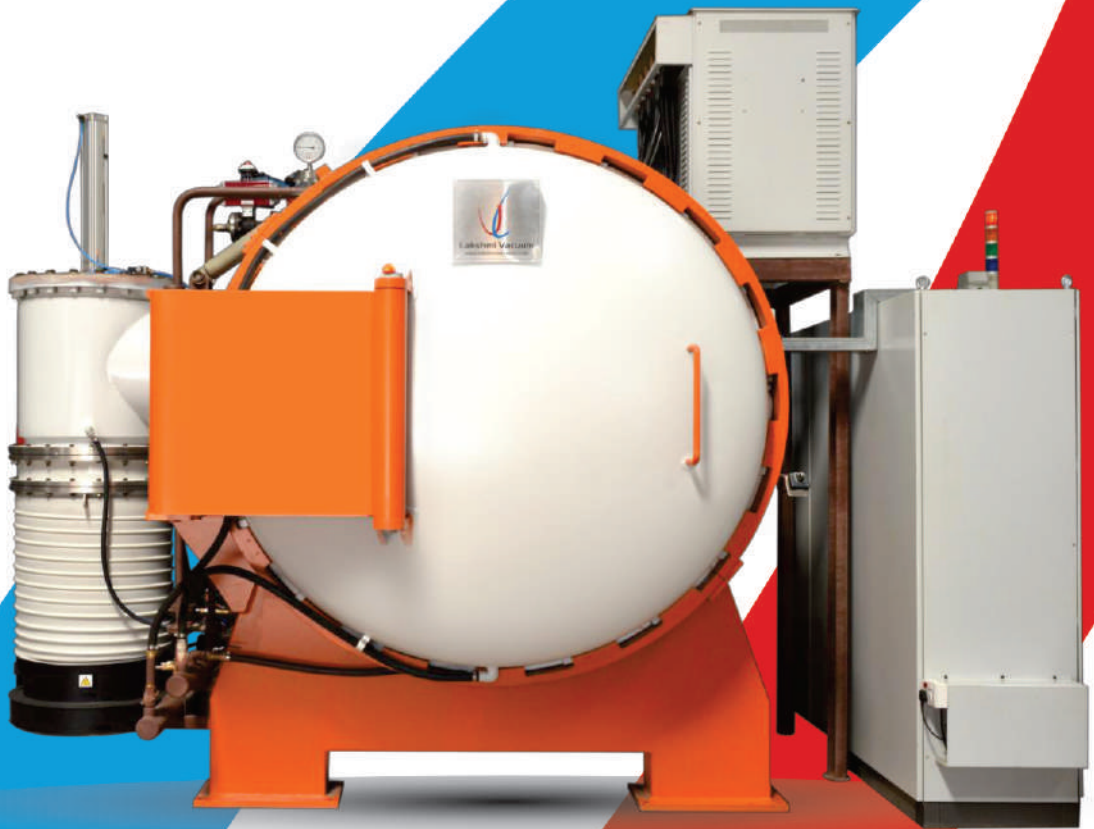


REVOLUTIONIZING VACUUM TECHNOLOGY

- VACUUM FURNACE MANUFACTURING
- VACUUM HEAT TREATMENT
- SPECIAL TOOL STEEL



LAKSHMI VACUUM TECHNOLOGIES PVT LTD

ABOUT US

Lakshmi Vacuum Group, founded in 2003, has grown into a leading entity in thermal processing, significantly impacting the Indian industrial sector. The group has demonstrated a commitment to excellence and innovation with a diverse portfolio encompassing Furnace Manufacturing, Vacuum Heat Treatment, and Distribution of Specialized Tool Steel. The Lakshmi Vacuum Group is anchored by its key entities, each contributing uniquely to its success.

KEY ENTITIES

Lakshmi Vacuum Heat Treaters Pvt Ltd:

- 🔧 Establishment: 2003
- 🔧 Presence: 8 Locations across India
- 🔧 Business Focus: Leading provider of Vacuum Heat Treatment services
- 🔧 Distinctive Features: Commitment to quality, widespread geographical coverage, and a customer-centric approach.

Lakshmi Vacuum Technologies Pvt Ltd:

- 🔧 Business Unit: Furnace Manufacturing
- 🔧 Brand Recognition: A reputed brand in India for Vacuum Furnaces
- 🔧 Specialization: State-of-the-art Vacuum Furnace Manufacturing
- 🔧 Achievements: Recognized for technological innovation and superior manufacturing practices.



DOMINATING FORCE IN INDIA: GEOGRAPHICAL PRESENCE

Nationally
recognized
leader in
**Vacuum
Technology**



LOCATIONS

Bangalore - 3 plants
IMT Manesar
Hosur
Coimbatore
Chennai
Hyderabad
Mumbai
Pune

STATISTICS & TRACK RECORD



400+
Furnace Manufactured



8
Heat Treatment Units



10000+
Client Base



Year in business

MANUFACTURING EXPERTISE OF LVTPL

1

SPECIFICATION

Process Capabilities Of Furnace

DETAILS

Quenching, Brazing, Carburizing, Debinding, Sintering, Surface Treatment, Annealing, Tempering.

2

SPECIFICATION

Special Process Capabilities

DETAILS

Vacuum Aluminium Brazing, Vacuum Nitriding, Aluminiuming Process.

3

SPECIFICATION

Temperature Parameters Of The Furnaces

DETAILS

As Per Ams 2750 Latest Rev
Class Of Uniformity: 1 & 2 (+/- 3 Deg C & +/- 5 Deg

4

SPECIFICATION

Leak Rate Of Furnaces

DETAILS

Conformance To Ams 2769 Latest Rev

5

SPECIFICATION

Pressure Vessel Design

DETAILS

As Per Asme Sec Code / Ce Conformance

VACUUM HEAT TREATMENT FURNACE (VHF SERIES)

- ✦ Lakshmi Vacuum is known for their VHF series of furnaces for thermal processing. Lakshmi Vacuum manufactures very dependable furnaces that are concerned with quality and Reliability.
- ✦ Engineered to perfection this series of furnaces are compliant to international standards Such as AMS & ASME.
- ✦ The VHF Series Vacuum Furnaces have low consumption of electricity, Gas and other utilities. And these machines are completely eco friendly.
- ✦ Lakshmi Vacuum puts a value addition to customers by providing hands on training in equipment and metallurgical services to customers enhancing the experience of the client in handling the equipment.
- ✦ The VHF series by Lakshmi Vacuum is one of the Highest sold Vacuum Furnaces in Indian Market.

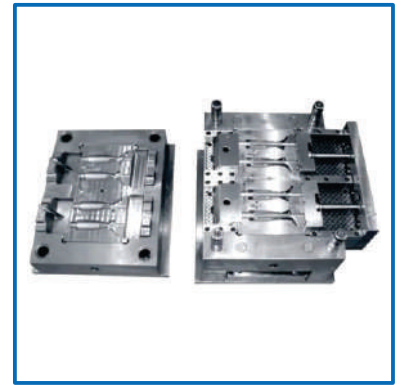


FEATURES

- ✦ Fully Automated Process (PLC + Scada)
- ✦ Selection of temperature range depending on process
- ✦ Having flexibility to process multiple requirements
- ✦ No Geometrical structure limit for heat treatment
- ✦ Very less consumption of utilizes (Gas, Water and Power)
- ✦ Repeatable Results
- ✦ High and Fine Vacuum Facility depending on the process requirements
- ✦ Multi Zone /Single zone temperature control
- ✦ Proven Cooling Efficiency

APPLICATION

1. Quench Hardening of All Tool Steel
2. Vacuum Annealing
3. Vacuum Stress Relieving
4. Through Hardening (oil & gas Quench)



TECHNICAL DATA SHEET

VHF SERIES FURNACE	VHF 400	VHF 600	VHF 900
Work Zone Size	400x400x600 mm	600x600x900 mm	900x900x1200 mm
Weight Capacity	300 Kg	600 Kg	1000 Kg
Operating Vacuum	10 ⁻² mbar (10 ⁻⁶ mbar optional)	10 ⁻² mbar (10 ⁻⁶ mbar optional)	10 ⁻² mbar (10 ⁻⁶ mbar optional)
Operating Temp.	1250° C	1250° C	1250° C
Quench Pressure	6/12/15/20 or Oil Quench	6/12/15/20 or Oil Quench	6/12/15/20 or Oil Quench
Gas Flow Pattern	N360°	N360°	N360°
Process Control	PLC & SCADA with IPC	PLC & SCADA with IPC	PLC & SCADA with IPC

* Customization as per Client Requirement

VACUUM BRAZING FURNACE (VBF SERIES)

- ✚ Lakshmi Vacuum is known for their VBF series (brazing) of furnaces for thermal processing. Lakshmi Vacuum manufactures very dependable furnaces that are concerned with quality and Reliability.
- ✚ Engineered to perfection this series of furnaces are compliant to international standards Such as AMS 2750 Latest Rev
- ✚ The VBF Series Vacuum brazing Furnaces have low consumption of electricity, Gas and other utilities. And these machines are completely eco friendly
- ✚ Lakshmi Vacuum puts a value addition to customers by providing hands on training in equipment and metallurgical services to customers enhancing the experience of the client in handling the equipment
- ✚ The VBF series by Lakshmi Vacuum is one of the Highest sold Vacuum Brazing Furnaces in Indian Market

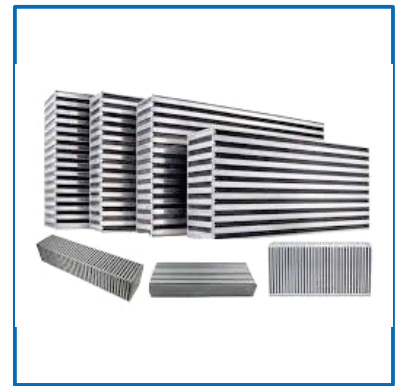
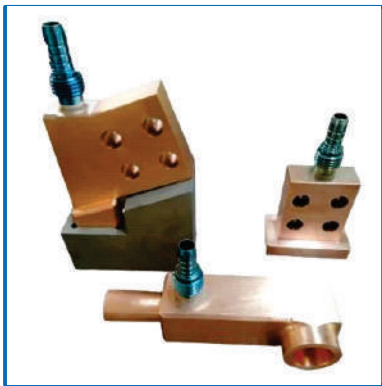


FEATURES

- ✚ Fully Automated Process (PLC + Scada)
 - ✚ Selection of temperature range depending on process
 - ✚ Having flexibility to process multiple requirements
 - ✚ No Geometrical structure limit for heat treatment
 - ✚ Very less consumption of utilizes (Gas, Water and Power)
 - ✚ Repeatable Results
 - ✚ High and Fine Vacuum Facility depending on the process requirements
 - ✚ Multi Zone /Single zone temperature control
 - ✚ Proven Cooling Efficiency
 - ✚ Power management control system
 - ✚ Orientation Options (Horizontal / Vertical loading)
-

APPLICATION

1. Metal to Metal Brazing
2. Carbide to Metal Brazing
3. Radiator Brazing
4. Aluminium Heat Exchanger Brazing
5. Vane Brazing
6. Aero Space Application
7. Automobile Defence and Medical



TECHNICAL DATA

VBF SERIES FURNACE	VBF 400	VBF 600	VBF 900
Work Zone Size	400x400x600 mm	600x600x900 mm	900x900x1200 mm
Weight Capacity	300 Kg	600 Kg	1000 Kg
Operating Vacuum	10 ⁻² mbar	10 ⁻² mbar	10 ⁻² mbar
High Vacuum Option	(10 ⁻⁶ mbar optional)	(10 ⁻⁶ mbar optional)	(10 ⁻⁶ mbar optional)
Operating Temp.	1250° C	1250° C	1250°C
Quench Pressure	2/12 bar	2/12 bar	2/12 bar
Gas Flow Pattern	Nozzle design 360° towards job	Nozzle design 360° towards job	Nozzle design 360° towards job
Process Control	PLC & SCADA with IPC	PLC & SCADA with IPC	PLC & SCADA with IPC

* Note For Vacuum Aluminium Brazing The Size And Capacity Differ With Respect To Requirement

VACUUM CARBURIZING FURNACE (PRECI CAB -SERIES)

- ✦ Lakshmi Vacuum is known for their VCF series (carburizing) of furnaces for thermal processing. Lakshmi Vacuum manufactures very dependable furnaces that are concerned with quality and Reliability.
- ✦ Engineered to perfection this series of furnaces are compliant to international standards Such as AMS 2750 Latest Rev
- ✦ The VCF Series Vacuum Carburizing Furnaces have low consumption of electricity, Gas and other utilities. And these machines are completely eco friendly
- ✦ Lakshmi Vacuum puts a value addition to customers by providing hands on training in equipment and metallurgical services to customers enhancing the experience of the client in handling the equipment.



ADVANTAGES

1. Minimization of Heat Treatment Distortion.
 2. Capability of High temperature Carburizing.
 3. Shorter Cycle Times.
 4. No Inter Granular Oxidation.
 5. Complex Geometrical Parts can be processed
 6. Eco-Friendly Process.
-

APPLICATION

1. Carburizing
2. Carbo-nitriding
3. Through Hardening
4. Carburizing Hardening
5. Carburizing Annealing
6. Tool Steel Hardening
7. Tempering
6. Annealing
7. Stress Relieving



TECHNICAL DATA

PRECI CARB SERIES	PRECI CARB 400 GQ/ OQ	PRECI CARB 600 GQ / OQ	PRECI CARB 900 GQ / OQ
WORK ZONE SIZE	400x400x600 mm	600x600x900 mm	900x900x1200 mm
WEIGHT CAPACITY	300 Kg	600 Kg	1000 Kg
OPERATING VACUUM	10 ⁻² mbar (10 ⁻⁶ mbar optional)	10 ⁻² mbar (10 ⁻⁶ mbar optional)	10 ⁻² mbar (10 ⁻⁶ mbar optional)
OPERATING TEMP.	1250° C	1250° C	1250°
QUENCH OPTION GAS	12/15/20	12/15/20	12/15/20
QUENCH OPTION OIL	Oil Quench with Agitation	Oil Quench with Agitation	Oil Quench with Agitation
PROCESS CONTROL	PLC & SCADA with IPC	PLC & SCADA with IPC	PLC & SCADA with IPC

VACUUM SINTERING FURNACE (VSF -SERIES)

- ✚ Lakshmi Vacuum is known for their VSF series (Sintering) of furnaces for thermal processing. Lakshmi Vacuum manufactures very dependable furnaces that are concerned with quality and Reliability.
- ✚ Engineered to perfection this series of furnaces are compliant to international standards
- ✚ Such as AMS 2750 E.
- ✚ The VSF Series Vacuum sintering Furnaces have low consumption of electricity, Gas and other utilities.
- ✚ And these machines are completely eco-friendly.
- ✚ Lakshmi Vacuum puts a value addition to customers by providing hands on training in equipment and metallurgical services to customers enhancing the experience of the client in handling the equipment
- ✚ VSF series of furnaces come with De – binding and non-De-binding Facility also.

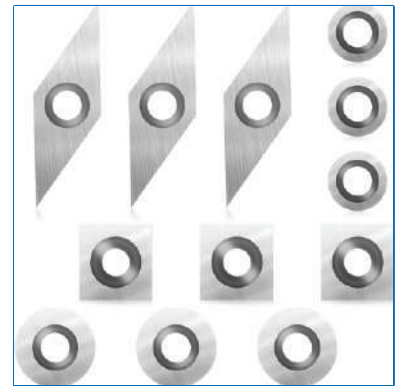
ADVANTAGES

1. Furnaces has options of Non-Metal and Metallic Hot Zone Compatibility.
2. High Sintering Temperature of up to 2500 Deg C
3. Vacuum levels to 10^{-6} Torr (8×10^{-5} mbar)
4. Partial pressure with argon, nitrogen or hydrogen for de-binding
5. Partial pressure range of 50 to 1000 microns (argon/nitrogen)
6. Fast cooling of up to 2 bar.
7. Binder removal system.
8. Burn-off system provided for safe handling of any flammable by-products



APPLICATION

1. Powder Metallurgy
2. 3 D Printing Application
3. Sintering of Carbide Tools
4. Sintering of Ceramics
5. Metal Injection Moulding



TECHNICAL DATA

VSF SERIES	VSF 400	VSF 600	VSF 900
Work Zone Size	400x400x600 mm	600x600x900 mm	900x900x1200 mm
Weight Capacity	300 Kg	600 Kg	900 Kg
Operating Vacuum	10 ⁻² mbar (10 ⁻⁶ mbar optional)	10 ⁻² mbar (10 ⁻⁶ mbar optional)	10 ⁻² mbar (10 ⁻⁶ mbar optional)
Operating Temp.	1250° C- 2500 Deg C	1250° C-2500 Deg C	1250°C-2500 Deg
Quench Pressure	6/12/15/20 bar	6/12/15/20N	6/12/15/20N
Gas Flow Pattern	N360°	N360°	N360°
Process Control	PLC & SCADA with IPC	PLC & SCADA with IPC	PLC & SCADA with IPC

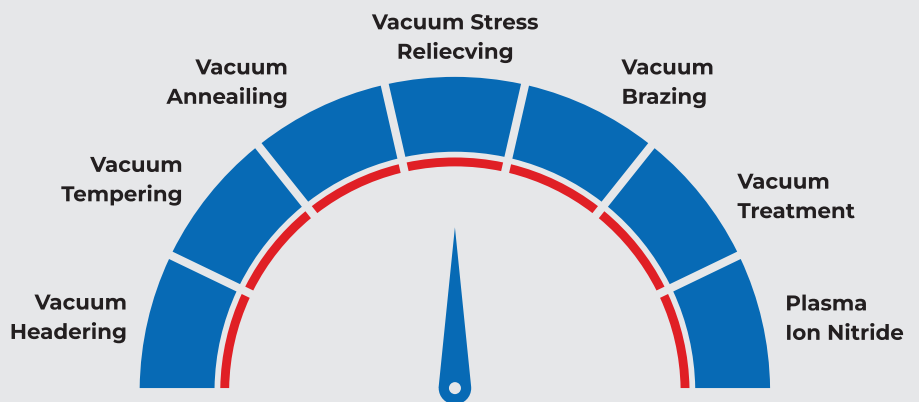
LAKSHMI VACUUM HEAT TREATERS PVT LTD



The Vacuum Heat Treatment Services is a division of Lakshmi Vacuum, which is strategically located in 8 Major Industrial Hubs in India. Heat Treatment Services are primarily aimed at customers who do not wish to invest in their own facilities, but who wish to benefit from all the well-known and repeatedly demonstrated advantages of the heat treatment technology developed by Lakshmi Vacuum. Our service centre has world Class Facilities in order to be able to offer customers optimized solutions even for complex heat treatment issues. We are one of the largest Vacuum Heat Treatment Service Providers Pan India.

Our main services are: Hardening in vacuum, High Pressure Gas Quenching (HPGQ) with nitrogen or Argon, Nitriding, Vacuum Brazing at high temperatures, Annealing in vacuum, Cryogenic Treatment, Vacuum Carburizing.

PROCESSES TREATMENT



FEATURES

- No scaling or decarburization
- Bright surface finish
- Uniform & gradual heating
- Uniform temperature throughout the hot zone
- Optimum process know-how ensures adequate soaking times.

INDUSTRIES



Aerospace



Automobile



Defence



Electrical &
Electronics



Commercial
Heat Treatment



IMT MANESAR



COIMBATORE PLANT



CHENNAI PLANT



MUMBAI



BENGALURU



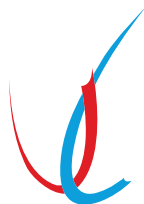
HYDERABAD



HOSUR



PUNE PLANT



Lakshmi Vacuum

LAKSHMI VACUUM TECHNOLOGIES PVT LIMITED

Unit - I No. 478 Sompura 2nd Stage, Industrial Area,
Thymagondlu Nelamanagala Taluk, Bangalore Rural - 562111, India.

Corporate Office

3, B-05, Ground floor, MS Industrial Complex,
14th Cross, Peenya Industrial Area, Bangalore - 560 058, Karnataka, India.
E-mail: marketing@lakshmivacuum.com Ph: +91 70 22 038 671

LAKSHMI VACUUM HEAT TREATERS PVT. LTD,

Lakshmi Vacuum Heat Treaters Pvt. Ltd,
NO 27 & 28 , 5th Main , 3rd Phase,
Peenya Industrial Area, **BANGALORE**-560058
+91 92 43 43 1484

Lakshmi Vacuum Heat Treaters
Office : #B/46, Balaji Industrial Estate,
Opp. Mahindra & Mahindra Yellow Gate,
Kandivali (East), **MUMBAI** - 400 101

Plant : #3, Landmark Industrial Estate,
Behind Union Bank of India,
Near Jivdani Hotel, Waliv Phata,
VASAI (E) - 401 208, Dist. Palghar,
MAHARASHTRA
022 20 89 5902 / +91 98 20 790060

Lakshmi Vacuum Metallurgicals Pvt. Ltd,
New Address : Plot No PAP – V- 45 , MIDC, Chakan
Industrial Area, Phase – 2, Wasuli, Taluka Khed,
Dist- **PUNE** – 410501.
+91 91 68 783 636

Lakshmi Vacuum Heat Treaters Pvt. Ltd,
Gurugram
Plot No 412D, Sector-7, IMT MANESAR,
Gurugram (HR) - 122050
+91 99 99 10 4469

Lakshmi Vacuum Heat Treaters Pvt. Ltd,
Shed No.D43/B, Phase IV Extention,
IDA Jeedimetla, **HYDERABAD** - 500 055
+91 92 46 879 845

Lakshmi Vacuum Services Pvt. Ltd, Chennai
Plot No: S 8, SIPCOT Aerospace Park, Vallam Vadagal,
Sriperumbudur, **KANCHIPURAM** Dist 602105.
+91 92 82 125 001

Lakshmi Vacuum Heat Treaters Pvt. Ltd,
Coimbatore
#464/2, Varathiyangarpalayam,
Kondayampalayam Post, Sarkarsamakulam
Village, **COIMBATORE** – 641 110
+91 99 80 562 722

Lakshmi Vacuum Heat Treaters Pvt. Ltd,
Hosur
No.1/406-17, SLV Nagar, Zuzuvadi,
Sidco 3rd Phase, **HOSUR** - 635 126 Tamil Nadu,
+91 82 20 001 486